

Machine Learning Assessment

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Spotter

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Freight Rate Prediction

1. Why is ML used and why does it beat a fixed rule?

Not every data analysis task requires an ML model to get a useful insight or answer.

In this case the fixed rule to predict a freight rate was not the optimal approach since the dataset presented had multiple numerical variables related to a freight. Fixed thresholds based on variables like distance and weight could have had certain effectiveness on the analysis being done; nevertheless, the creation of said thresholds would require knowledge on the freight shipping industry and its complicated and costly to have accountability of every variable that affects rates, in that sense a machine learning model can facilitate those decisions and learn patterns that human analysis could not take into account to predict future rates.

2. Approach and Validation process

For creating a reliable ML model the first step is setting the data contract and understanding the datasets and the variables related. First it was understood what one row meant and how every variable was related to the target feature. The validation strategy was to see what model extrapolates better to unseen future data and how well it fits the data it was trained on. The split was a time-based split where data was divided chronologically on two sections. The first section was the “Train” section from January 1 to August 31, 2025 and the “Holdout” section from September 1 to October 31, 2025. This split is not random since a standard 80/20 split would scatter rows from multiple months so the model could learn from data from the training and validation datasets, therefore the model would see different data from the same months therefore a data leakage could occur and the model would not have the capacity of accurate extrapolation. The time based split forecasts a real scenario at smaller scale using the holdout section and based on the performance and results on that holdout section the extrapolation on November and December is set to have accurate extrapolation values.

The feature engineering done uses data cleaning statistics like using the median in the numerical variables and carefully used in the training and applied unchanged to the holdout

section. Moreover, no data derived from outside training was used on how the training data was prepared. The next step was to compare models using the “sklearn” library and see their effectiveness on the data, Linear Regression and Gradient Boosted Trees where the models to be compared. These models were trained on the split mentioned before. Even though GBM scored higher metrics, it had problems generating the December chart. The reason being that tree-based models can not extrapolate a continuous future like the cyclical modification made to the date variable to correctly estimate the calendar year. On the other hand linear regression extrapolates trend and cyclical seasonality, therefore a better representation on the December chart was achieved, the final decision was to use linear regression model.

3. December Chart

